

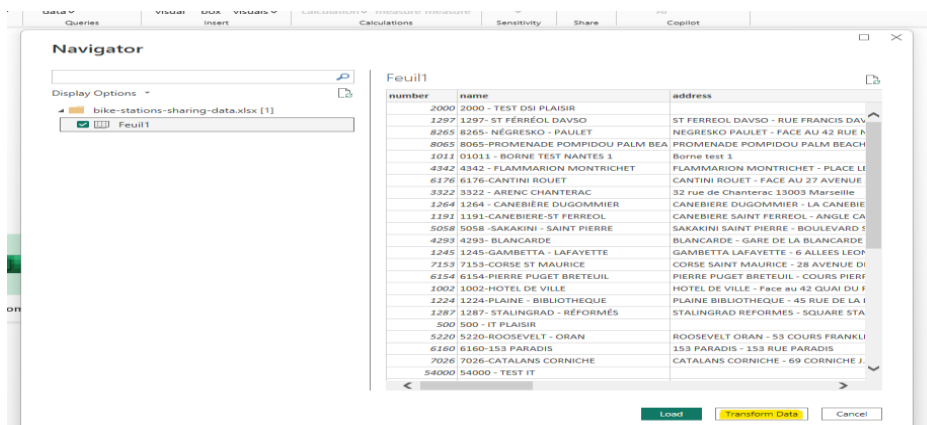
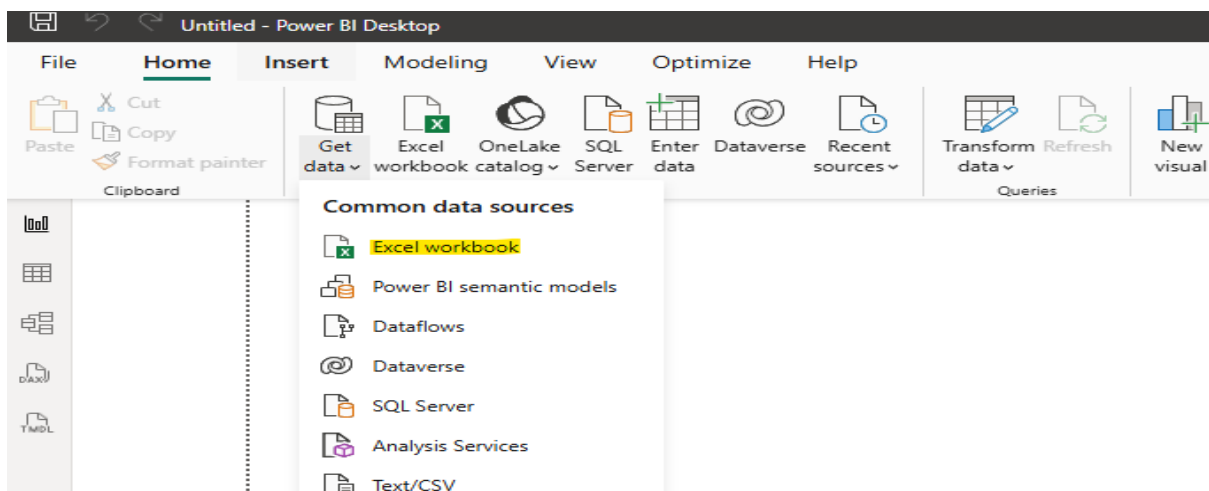
Analyzing Smart City Bike Sharing Data using Power BI

Data Import & Transformation

Loading of Dataset:

1. Open Power BI – In Home -> Choose excel workbook in data tab
2. In the open dialogue box -> Select the file you want to load -> Click Open -> Select Transform Data

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The screenshot shows the Power Query Editor with the 'Transform' tab selected. The formula bar contains the M code: `= Table.TransformColumnTypes(#"Promoted Headers",{{"number", Int64.Type}, {"name", type text}, {"address", type text}})`. The data table below has three columns: 'number' (type 1.2), 'name' (type ABC), and 'address' (type ABC). Each column has a validity status bar showing 100% Valid, 0% Error, and 0% Empty. The data rows are as follows:

number	name	address
2000	2000 - TEST DSI PLAISIR	
1297	1297- ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL
8265	8265- NÉGRESKO - PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO
8065	8065-PROMENADE POMPIDOU PALM ...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...
1011	01011 - BORNE TEST NANTES 1	Borne test 1
4342	4342 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...
6176	6176-CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI

Identifying Data Types:

Changed the data type of position(longtitude/latitude) column from text(ABC) to decimal number(1.2)

The screenshot shows the 'position' column in the data table. The column header is 'position' (type ABC). The validity status bar shows 95% Valid, 0% Error, and 5% Empty. The data rows are as follows:

position
46.5488766782771, 15.6350265324683
43.293485736186526, 5.378423799723321

Changed the data type of last update column to Date/Time from date/Tim/Zone. This was changed because all entries has same time zone

The screenshot shows the 'Last Update' column in the data table. The column header is 'Last Update' (type Date/Time). The validity status bar shows 100% Valid, 0% Error, and 0% Empty. The data rows are as follows:

Last Update
02-06-2022 15:14:38 +05:30
06-04-2022 12:54:35 +05:30
27-09-2022 10:57:48 +05:30
08-06-2022 11:50:13 +05:30
23-05-2022 20:11:50 +05:30
13-09-2022 11:28:01 +05:30

Cleaning the data – Power Query Transformations:

- **Standardizing the position column:**
 1. Select the position column and follow the below steps.
 2. Select split column icon from Transform tab

3. Select delimiter by comma -> click OK
4. Rename the separated columns into Latitude and Longitude

The screenshot shows the Power Query Editor interface. On the left, the 'Split Column by Delimiter' dialog box is open, showing the 'Split at' section with 'Each occurrence of the delimiter' selected. The 'Advanced options' section shows 'Quote Character' set to 'None'. The main area displays a data table with columns for 'address', 'position', and 'banking'. The 'position' column is highlighted, and the 'Split Column by Delimiter' dialog box is open over it. The resulting data table shows the 'position' column split into 'Latitude' and 'Longitude' columns. The 'address' column contains various street names, and the 'banking' column contains 'Valid', 'Error', or 'Empty' values.

address	Latitude	Longitude	banking
AVSO ANGLE RUE SAINT FERREOL	46.5488766782771	15.6350265324683	Valid
E NEGRESKO	43.293485736186526	5.378423799723321	Valid
CH - 6 PROMENADE GEORGES PO...	43.269318556371154	5.389387542314263	Valid
	43.267272515671856	5.372088524255505	Valid
	47.19504	-1.5575	Valid

• Handling Missing values

Column Address:

The column address has null values, where we can find it through column quality filed in View tab.

The screenshot shows the Power Query Editor interface. The 'View' tab is selected, and the 'Column quality' checkbox is checked. The main area displays a data table with columns for 'Latitude', 'Longitude', and 'address'. The 'address' column is highlighted, and the 'Column quality' dialog box is open over it. The resulting data table shows the 'address' column with various street names, and the 'Latitude' and 'Longitude' columns with their respective values. The 'address' column contains null values, which are highlighted in yellow.

Latitude	Longitude	address
46.5488766782771	15.6350265324683	AVSO ANGLE RUE SAINT FERREOL
43.293485736186526	5.378423799723321	E NEGRESKO
43.269318556371154	5.389387542314263	CH - 6 PROMENADE GEORGES PO...
43.267272515671856	5.372088524255505	
47.19504	-1.5575	

[illegible]

{"",null,"Not Available",Replacer.ReplaceValue,{"address"})			
	A ^B C address	1.2	Latin
100%	Valid	100%	Valid
0%	Error	0%	Error
0%	Empty	0%	Empty
	Not Available		
	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL		
	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO		
LM ...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...		
	Borne test 1		
IT	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...		
	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI		
	32 rue de Chanterac 13003 Marseille		

- Column Name and Address:**

In order to make the name and address column consistent, used format -> UPPER which in turn changes every text in the selected column to upper case.

TransformColumns("#Replaced null as NA in address column",{ "name", Text.Upper, type text}

	A ^B C name		A ^B C address		1.
6	Valid 100%		Valid 100%		
6	Error 0%		Error 0%		
6	Empty 0%		Empty 0%		
70	2000 - TEST DSI PLAISIR		NOT AVAILABLE		
97	1297- ST FÉRRÉOL DAVSO		ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL		
55	8265- NÉGRESKO - PAULET		NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO		
55	8065-PROMENADE POMPIDOU PALM ...		PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...		
11	01011 - BORNE TEST NANTES 1		BORNE TEST 1		
42	4342 - FLAMMARION MONTRICHET		FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...		
76	6176-CANTINI ROUET		CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI		

Then, I have added custom column named as “Station Name” to standardize the name column by using the below syntax. I have removed the station id as it was already available as separate column

Syntax:

Text.Trim(Text.Replace(Text.TrimStart(Text.Replace([name],Text.From([number]), ""),{"-",";",""}),"-"," "))

	Last Update	ABC 123 Station Name
100%	Valid 100%	Valid 100%
0%	Error 0%	Error 0%
0%	Empty 0%	Empty 0%
0	02-06-2022 15:14:38	TEST DSI PLAISIR
0	06-04-2022 12:54:35	ST FÉRRÉOL DAVSO
0	27-09-2022 10:57:48	NÉGRESKO PAULET
0	08-06-2022 11:50:13	PROMENADE POMPIDOU PALM B...
0	23-05-2022 20:11:50	0 BORNE TEST NANTES 1
0	13-09-2022 11:28:01	FLAMMARION MONTRICHET
0	06-09-2022 18:02:17	CANTINI ROUET
7	14-09-2022 13:27:09	ARENC CHANTERAC
0	12-09-2022 11:18:28	CANEBIÈRE DUGOMMIER

Column Contract Name:

This column has some inconsistency in data, therefore used Capitalize each Word function to capitalize the first letter

	ABC Contract Name		123 Bike Stars
100%	Valid	100%	Valid
0%	Error	0%	Error
0%	Empty	0%	Empty
	maribor		
	marseille		
	marseille		
	marseille		
	nantes		
	marseille		
	marseille		

	A ^B _C Contract Name		1 ² ₃ Bike Stands
100%	Valid	100%	Valid
0%	Error	0%	Error
0%	Empty	0%	Empty
	Maribor		
	Marseille		
	Marseille		
	Marseille		
	Nantes		
	Marseille		
	Marseille		
	Marseille		

Creating Custom Columns:

Banking Status:

Created conditional column named “Banking Status” using the below IF syntax to categorize the true/false as banking available/no banking.

Empty

EN	Maribor	1	0	0	02-06-2022 15:14:38
EN	Marseille	9	0	0	06-04-2022 12:54:35
EN					28-11-2022 14:00:36

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Banking Status

Column Name	Operator	Value	Output
If banking	equals	ABC 123 TRUE	Then ABC 123 Banking Available

Add Clause

Else
ABC 123 No Banking

OK Cancel

EN	Nancy	1	1	0	28-11-2022 14:00:36
----	-------	---	---	---	---------------------

seille	9	0	0	06-04-2022 12:54:35	ST FÉRRÉOL DAVSO
--------	---	---	---	---------------------	------------------

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Bonus Status

Column Name	Operator	Value	Output
If bonus	equals	ABC 123 TRUE	Then ABC 123 Bonus Station

Add Clause

Else
ABC 123 Regular Station

OK Cancel

cy	1	1	0	28-11-2022 14:00:36	TEST IT
----	---	---	---	---------------------	---------

	ABC 123 Banking Status	ABC 123 Bonus Status
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
	No Banking	Regular Station
SO	Banking Available	Regular Station
LET	Banking Available	Regular Station
MPIDOU PALM B...	Banking Available	Regular Station
IANTES 1	No Banking	Regular Station
ONTRICHET	Banking Available	Regular Station
	Banking Available	Regular Station
AC	Banking Available	Regular Station
OMMIER	Banking Available	Regular Station
RREOL	Banking Available	Regular Station
PIERRE	Banking Available	Regular Station

Changing all the Header names for easy reference:

Data Modelling:

Main Fact table:

Renamed my main data set table to Fact_BikeStationStatus where I cleaned and transformed all data

Creating Dimension Tables:

DimStation:

Steps

1. Right-click Fact_BikeStationStatus
2. Select Reference (not Duplicate)
3. A new query is created. Rename it as DimStation
4. Keep only these columns:

- Station ID
- Station Name
- Station Address
- Latitude
- Longitude
- Banking Status
- Bonus Status

5. Remove duplicates by following below steps:

6. Home → Remove Rows → Remove Duplicates using: Station ID

This ensures one row per station.

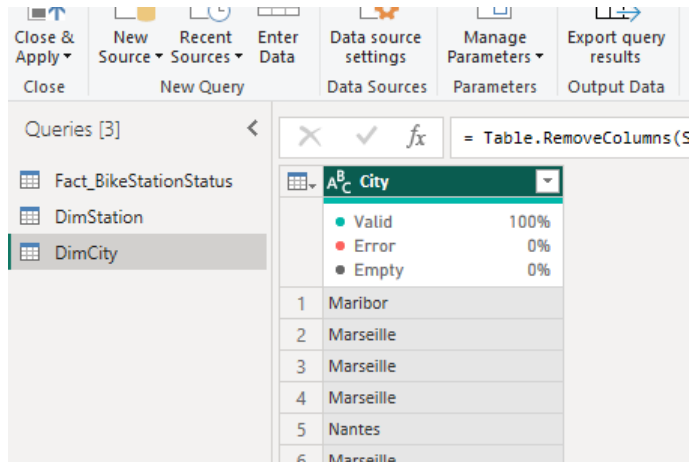
Close & Apply	New Source	Recent Sources	Enter Data	Data source settings	Manage Parameters	Export query results	Refresh Preview	Manage	Choose Columns	Remove Columns	Keep Rows	Remove Rows
Close	New Query			Data Sources	Parameters	Output Data		Query	Manage Columns		Reduce Rows	

Queries [2]			
Fact_BikeStationStatus			
DimStation			

= Table.Distinct(#"Removed Columns", {"Station ID"})			
1.2	Station ID	A. Station Address	1.2 Latitude
	Valid 100%	Valid 100%	Valid
	Error 0%	Error 0%	Error
	Empty 0%	Empty 0%	Empty
1	2000	NOT AVAILABLE	
2	1297	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL	
3	8265	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	
4	8065	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...	
5	1011	BORNE TEST 1	
6	4342	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...	
7	6476	CANTIN ROUET - FACE AU 37 AVENUE RUE CANTIN	

Create DimCity

1. Again Right-click Fact_BikeStationStatus
2. Select Reference
3. Rename it as DimCity
4. Keep only: Column City
5. Remove duplicates.



Creating DimDate

Go to: **Modeling** → **New Table**

Use below DAX:

```
DimDate = CALENDAR(MIN(Fact_BikeStationStatus[Update Date]),MAX(Fact_BikeStationStatus[Update Date]))
```

Add these columns to DimDate

Year

```
Year = YEAR(DimDate[Date])
```

Month Number

```
Month Number = MONTH(DimDate[Date])
```

Month

```
Month = FORMAT(DimDate[Date], "MMMM")
```

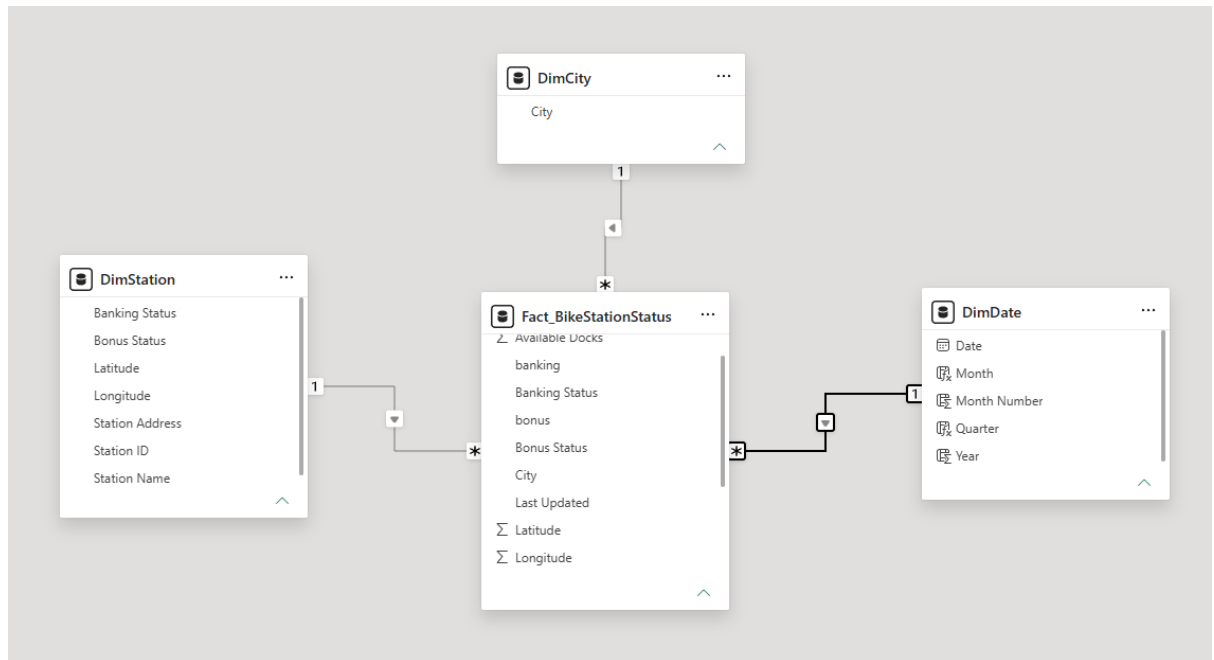
Quarter

```
Quarter = "Q" & FORMAT(DimDate[Date], "Q")
```

Clipboard	data	workbook catalog	server	data	sources	data	relationships	measure	measure column
Clipboard	Data				Queries	Relationships	Calculations		
	1 DimDate = CALENDAR(MIN(Fact_BikeStationStatus[Last Updated]),MAX(Fact_BikeStationStatus[Last Updated]))								
	Date	Year	Month Number	Month	Quarter				
	18 January 2022	2022	1	January	Q1				
	19 January 2022	2022	1	January	Q1				
	20 January 2022	2022	1	January	Q1				
	21 January 2022	2022	1	January	Q1				
	22 January 2022	2022	1	January	Q1				
	23 January 2022	2022	1	January	Q1				
	24 January 2022	2022	1	January	Q1				
	25 January 2022	2022	1	January	Q1				
	26 January 2022	2022	1	January	Q1				
	27 January 2022	2022	1	January	Q1				

Creating Relationships:

Star Schema has been used which has main fact table and their dimension tables connected like a star



Created Calculated Columns:

Occupancy %

Occupancy % = $\text{DIVIDE}([\text{Available Bikes}], [\text{Total Docks}])$

Empty Station

Empty Station = $\text{IF}([\text{Available Bikes}] = 0, "Yes", "No")$

Full Station

Full Station = $\text{IF}([\text{Available Bike Stands}] = 0, "Yes", "No")$

Utilization %

Utilization % = $\text{DIVIDE}([\text{Total Docks}] - [\text{Available Docks}], [\text{Total Docks}])$

Occupancy %	Empty Station	Full Station	Utilization %
0.05	No	No	0.1
0.1	No	No	0.1
0.65	No	No	0.65
0.45	No	No	0.9
0.05	No	No	0.05
0.9	No	No	0.95
1	No	Yes	1
0.4	No	No	0.4
0	Yes	No	0
0.35	No	No	0.35
0.05	No	No	0.15
0.95	No	Yes	1

Created Dax MEASURES:

Total Stations

Total Stations = DISTINCTCOUNT(Fact_BikeStationStatus[Station Id])

Total Bikes

Total Bikes =SUM(Fact_BikeStationStatus[Available Bikes])

Total Empty DocksTotal Empty Docks =SUM(Fact_BikeStationStatus[Available Docks])

Average Utilization

Average Utilization =AVERAGE(Fact_BikeStationStatus[Utilization %])

Open Stations

Open Stations
=CALCULATE(COUNTROWS(Fact_BikeStationStatus),Fact_BikeStationStatus[Station Status]="OPEN")

Closed Stations

Closed Stations

```
=CALCULATE(COUNTROWS(Fact_BikeStationStatus),Fact_BikeStationStatus[Station Status]="CLOSED")
```

Empty Stations Count

Empty Stations

```
=CALCULATE(COUNTROWS(Fact_BikeStationStatus),Fact_BikeStationStatus[Available Bikes]=0)
```

Dashboard Layout:

KPI Cards

Using the above DAX measures, created the below KPI's using cards by grading the DAX calculations to the card

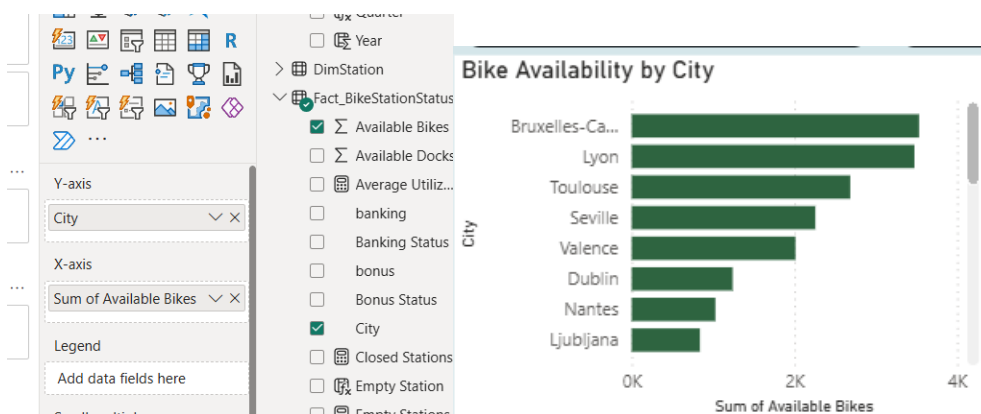
- Total Stations
- Total Bikes Available
- Open Stations
- Average Utilization %



Clustered Bar Chart

City vs Available Bikes

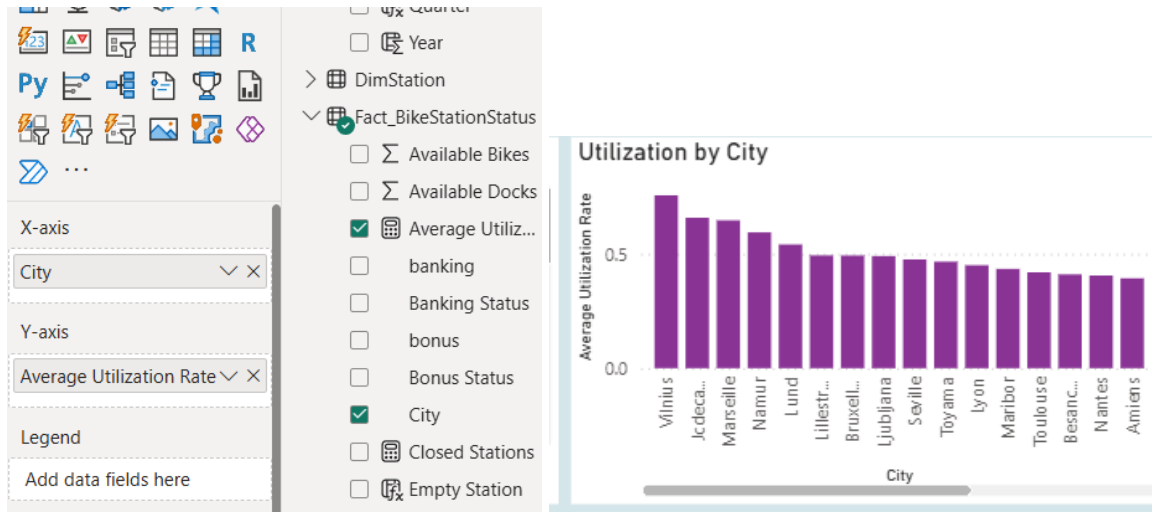
Shows cities with highest bike availability.



Column Chart

City vs Average Utilization %

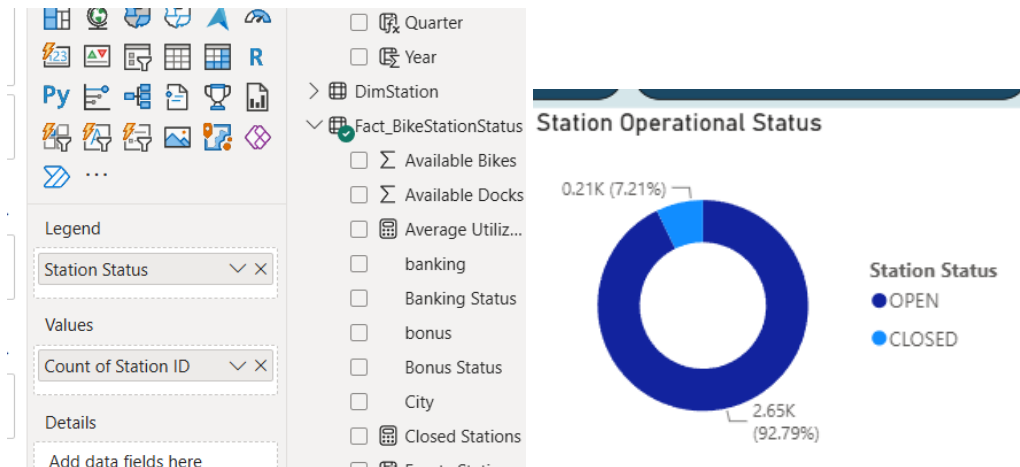
Find high-performing cities.



Donut Chart

Station Status

OPEN vs CLOSED

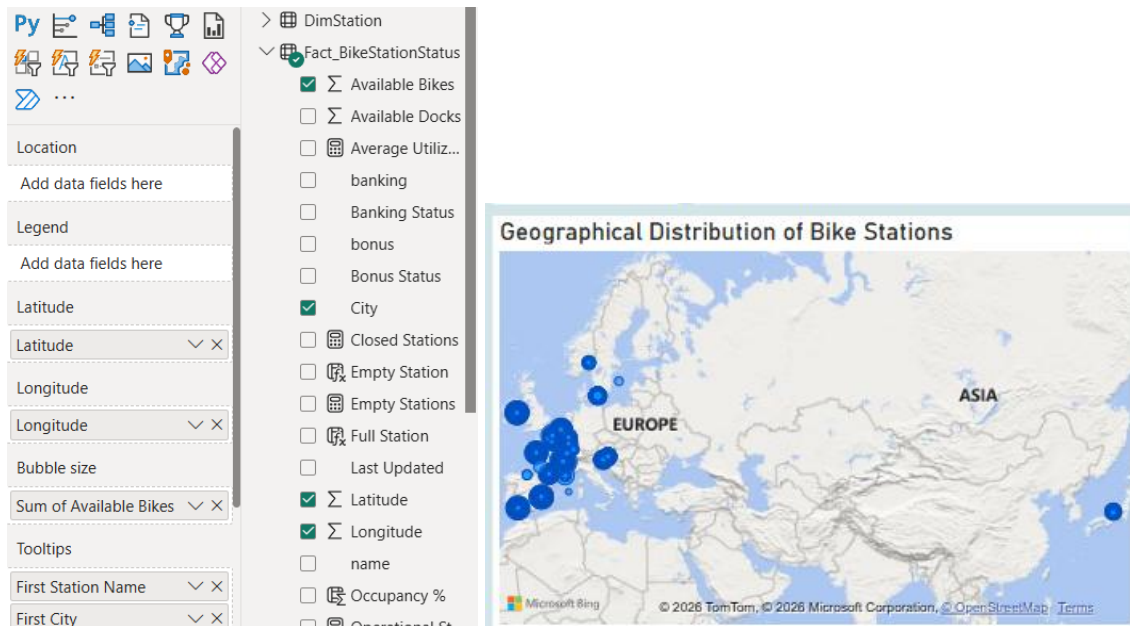


Map Visual

Latitude + Longitude

Size: Available Bikes

Shows station distribution.



Banking Status

Donut Chart

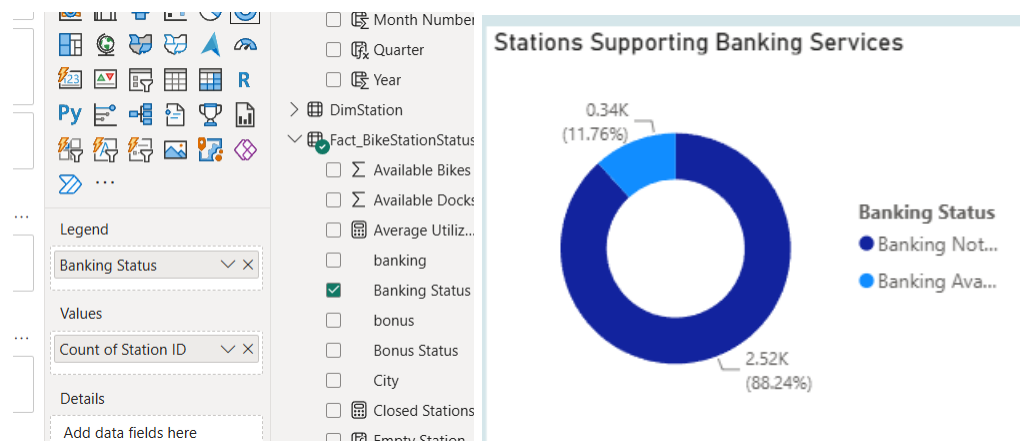
Fields:

Legend

Values

Banking Status

Count of Station ID



Stations Requiring Bike Replenishment

Add a Table Visual

1. Go to the Report View

2. In the Visualizations pane

3. Click Table

We will see an empty table appear on the canvas.

Add Columns to the Table

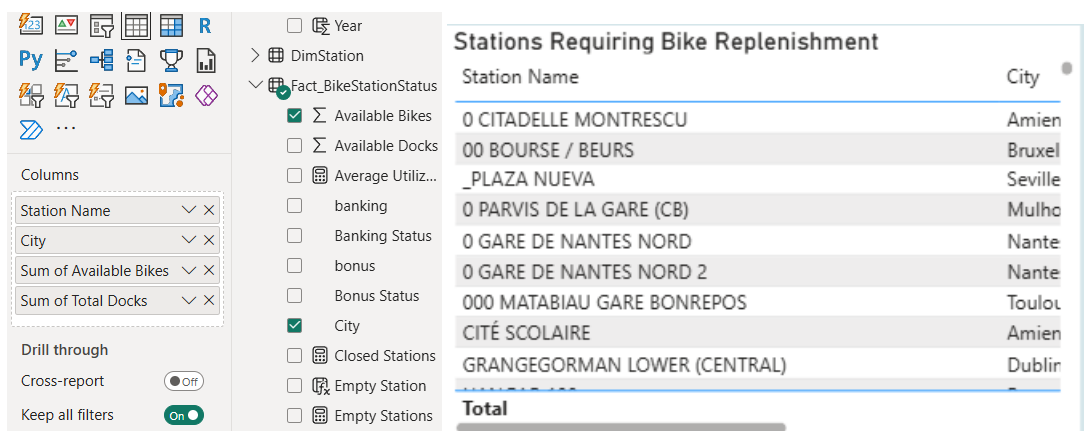
From the Fields pane, drag these fields into the table:

Station Name

City

Available Bikes

Total Docks



Dashboard:

